

Label Free Failure Detection for Land Cover Segmentation Across Continents and Sensors

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Abstract—A land cover segmentation model trained in one place is often deployed somewhere the training labels never covered, a different country, camera or season. This project trains a standard segmentation model on a large Chinese land cover dataset, then studies where and why it gets pixels wrong, along class boundaries, on rare classes, and where it is confidently wrong rather than visibly unsure. The main question is whether simple, label free scores computed from the model’s own output can flag which pixels and images are likely wrong, and whether that survives on imagery the model never trained on, including German cities on a different camera type and a worldwide benchmark spanning 44 countries on six continents. The scores do transfer, staying clearly better than a coin flip on every domain tested, though ranking is stronger on the two single city domains than the more diverse worldwide one. A simpler idea, one overall novelty score per image flagging an unfamiliar scene before any pixel is checked, does not work reliably, catching some new domains and missing others with no clear pattern. Full training, evaluation and scanning code is released.

I. INTRODUCTION

Semantic segmentation of aerial and satellite imagery feeds land cover mapping, disaster response and agricultural monitoring, where production imagery rarely matches the country, sensor or season a model trained on. A single overall mean IoU hides uneven failure across classes and unseen geography.

This study asks three questions. First, where and why does a fixed segmentation baseline fail, beyond one overall number. Second, can label free per pixel signals rank those failures without labels, and does that survive on geography, sensors and continents the model never trained on. Third, can a whole image novelty score serve as a reliable first pass alarm for an unfamiliar domain.

That pixel level uncertainty degrades less under distribution shift than raw accuracy has been shown before, in medical segmentation and driving scenes [1]–[4]. This paper tests the same question in remote sensing, across a sensor band change from RGB to infrared red green and a benchmark spanning dozens of countries and six continents, alongside a paired negative result, a whole image novelty score that does not show the same consistency.

The contributions are as follows.

- A reproducible DeepLabV3 ResNet50 baseline on LoveDA, trained independently twice with consistent scores, characterized along three failure axes, boundary against interior, rare against common classes, and confidence on correct against incorrect pixels.
- Four label free per pixel uncertainty signals evaluated in domain on LoveDA and cross domain on ISPRS Potsdam, ISPRS Vaihingen and OpenEarthMap, adding a sensor band change, a second European city and worldwide diversity.
- A paired evaluation of a whole image Mahalanobis novelty score as a domain alarm, showing it swings from below to above chance with no simple explanation.
- A label free scanning tool for arbitrary aerial imagery at native resolution, plus an interactive upload application.
- An open pipeline, training, evaluation, failure analysis and cross domain validation code, with exact data preparation commands.

II. RELATED WORK

Segmentation baselines. DeepLabV3 [5] with a ResNet [6] backbone is a strong, simple aerial segmentation baseline and is the model used here. LoveDA [7] spans urban and rural scenes across three Chinese cities, built to expose domain gaps. ISPRS Potsdam and Vaihingen [8] are German urban benchmarks differing from LoveDA in city, sensor and ground sample distance, and OpenEarthMap [9] tests worldwide generalization across 44 countries on six continents.

Label free uncertainty and misclassification detection. Maximum softmax probability [10] is the simplest label free failure signal and a strong baseline for out of distribution detection. Selective classification [11] formalizes trading coverage for risk using such a signal, motivating risk coverage area as a metric. Calibration work [12] explains why raw softmax confidence alone can be unreliable, motivating several signals rather than one.

Out of distribution and novelty detection. A Mahalanobis distance in feature space [13] flags inputs unlike the training distribution and is the basis of the whole image novelty score tested here, needing only one feature vector per image, attractive for a first pass alarm, though results here show that convenience is not free.

Uncertainty under domain shift. That label free uncertainty ranks errors well as accuracy drops under domain shift has precedent in digital pathology [1], driving scenes under simulation to real and weather shifts [2], [3], and medical vessel segmentation [4]. This study asks the same question in remote sensing land cover, adding a sensor band change and worldwide diversity, and reports where the complementary novelty tool breaks down.

III. MATERIALS AND METHODS

A. Baseline model and training

DeepLabV3 with a ResNet50 backbone, torchvision’s standard implementation, is used with the auxiliary classification head enabled and an ImageNet pretrained backbone. Augmentation is limited to horizontal and vertical flips and 90 degree rotations, keeping nadir aerial imagery geometrically valid without resampling artifacts. Training uses Adam, a cosine learning rate schedule decaying from $5e-4$ toward zero, cross entropy loss on both heads with the auxiliary head weighted at 0.4, mixed precision, and a 384 by 384 input, for 15 epochs. The checkpoint with the best validation mean IoU is kept. The model was trained independently twice, three weeks apart, to check the pipeline reproduces consistent scores.

B. Failure taxonomy

Given the trained model and labeled LoveDA validation split, the per class confusion matrix is computed and three views derived from it. A boundary view separates pixels near a class edge from interior pixels and reports mean IoU for each. A class size view groups classes into frequency bins and reports mean IoU and recall per bin. A confidence view compares mean maximum softmax probability on correct against incorrect pixels.

C. Label free per pixel signals

Four signals are computed from the raw logits with no ground truth involved. Confidence shortfall is one minus the maximum softmax probability. Predictive entropy is the Shannon entropy of the softmax distribution, normalized to zero to one. Margin risk is one minus the gap between the top two softmax probabilities. A blended failure score linearly combines normalized entropy, margin risk and confidence shortfall with weights 0.4, 0.4 and 0.2. Higher always means a higher estimated chance of being wrong.

D. Whole image novelty

A separate whole image score targets a model confidently and uniformly wrong on a scene unlike training, which a per pixel signal need not catch. Each image is represented by the global average pooled feature map from the backbone, a 2048 dimensional vector. A Gaussian, covariance shrunk toward its own diagonal, is fit once to 300 LoveDA training images. A new image is scored by Mahalanobis distance to that Gaussian, normalized into a zero to one novelty score. Since held out images score toward the high end of that range, every novelty comparison here measures a domain area under the curve between two held out runs rather than raw scores alone.

E. Label free scanning tool

A scanning tool slides a window across any input image at native resolution, preserving true ground sample distance, then stitches per window predictions and uncertainty maps into full size panels, writing a per image ranking so the least trustworthy images triage first with no labels. An interactive upload application exposes the same maps for a single image with a plain language novelty verdict. On six LoveDA validation tiles, per image mean failure score ranged from 0.01 on a uniform scene to 0.43 on the hardest agricultural mosaic, with the highest scoring regions falling along field boundaries and mixed settlement areas, the same regions the labeled taxonomy independently flags as weak.

F. Validation protocol

For a labeled split, every valid pixel is scored correct or wrong and treated as the target for a ranking problem, judged with the metrics in Section V, pixel level AUROC, AUPR, risk coverage and image level Spearman correlation. For novelty a domain AUROC is also computed, treating stored training distances as the negative class and the new domain's distances as the positive class.

IV. EXPERIMENTAL SETUP

A. Datasets

LoveDA [7] provides 5987 tiles at 0.3 meters per pixel across seven classes, background, building, road, water, barren, forest and agriculture, split into 2522 training and 1669 validation tiles, the latter used for the baseline and in domain validation. ISPRS Potsdam and Vaihingen [8] are six class benchmarks, impervious surface, building, low vegetation, tree, car and clutter, never used in training, obtained through a public mirror. Potsdam is German urban RGB imagery at 0.05 meters per pixel, 38 orthophotos of 6000 by 6000 pixels, cut into 2304 pixel patches so resizing to the 384 network input lands at 0.3 meters per pixel. Source photos split 30 training and 8 validation, yielding 120 training and 32 validation patches. Vaihingen is a different German city on an infrared, red, green sensor rather than RGB, at 0.09 meters per pixel, cut the same way, yielding 49 patches from all 33 tiles. OpenEarthMap [9] is an eight class benchmark, bareland, rangeland, developed space, road, tree, water, agriculture land and building, at 0.25 to 0.5 meters per pixel, spanning 44 countries on six continents. Its official validation split of 384 tiles is used directly, near the model input size, with no patch cutting. Its test split ships without labels in this mirror.

B. Cross domain class mapping

Potsdam and Vaihingen share a six class palette, coarsely mapped onto LoveDA's seven, impervious surface to road, building to building, low vegetation to background since LoveDA's agriculture means cropland rather than lawn, and tree to forest. Car and clutter have no counterpart and are excluded. OpenEarthMap's eight classes map with nothing excluded, bareland to barren, road to road, tree to forest, water to water, agriculture land to agriculture and building to building, while rangeland and developed space both join background. These mappings are deliberately coarse, absolute error rates on the foreign domains are approximate, though the ranking metrics only need the error mask roughly right.

V. EXPLANATION OF METRICS

This section defines, in plain language, every metric used to read the results in Section VI.

- **Mean IoU** (intersection over union) measures how well a predicted class region overlaps the true region, averaged across classes. 1.0 is a perfect match, 0.0 is no overlap.
- **Base error rate** is the fraction of pixels the model gets wrong on a split, the floor a random guess would compare against.

TABLE I: Per class validation results on LoveDA, main checkpoint.

Class	IoU	Precision	Recall	F1
Background	0.5273	0.6028	0.8080	0.6905
Building	0.6134	0.7103	0.8181	0.7604
Road	0.5403	0.8450	0.5997	0.7015
Water	0.6836	0.7779	0.8493	0.8120
Barren	0.2488	0.6065	0.2967	0.3985
Forest	0.3970	0.6135	0.5295	0.5684
Agriculture	0.5357	0.8815	0.5773	0.6977

- **AUROC** measures how well a label free score ranks wrong pixels above correct ones. 0.5 is a coin flip, 1.0 ranks every wrong pixel above every correct one, the main number used to judge whether a signal works.
- **AUPR** is a related ranking score whose chance level equals the base error rate rather than a fixed 0.5, useful since wrong pixels are usually the minority class.
- **Risk coverage** asks how low the error rate gets among only the pixels a signal trusts most, if the rest are discarded. Its area, AURC, summarizes the curve into one number, lower is better.
- **Spearman correlation** measures whether an image ranked riskier by the mean signal also has a higher true error rate. Near 1.0 is near perfect agreement, near 0.0 is no relationship.
- **Novelty domain AUROC** applies the same ranking idea at the dataset level, whether the novelty score alone tells a foreign domain's images apart from the training domain's.

VI. RESULTS

A. Baseline and failure taxonomy

The baseline reaches a mean IoU of 0.5066, pixel accuracy 0.6951 and mean F1 0.6613 on LoveDA validation, inside the published DeepLabV3 range. An earlier, independently trained run under the same configuration reached 0.5019 mean IoU, 0.6936 pixel accuracy and 0.6553 mean F1, closely matching and indicating the pipeline reproduces consistent scores. Table I breaks the main run down per class. Water is strongest at 0.6836 IoU, barren weakest at 0.2488, forest also weak at 0.3970.

The confusion matrix in Figure 1 shows a pattern beyond the per class scores. Background is over predicted, recall 0.8080 against precision 0.6028, while road and agriculture are under predicted, high precision (0.8450, 0.8815) alongside low recall (0.5997, 0.5773). The model plays it safe, preferring background over committing to a specific class, and misses real road and agriculture pixels more than it invents false ones.

Figure 2 shows the training dynamics. Training loss falls smoothly across all 15 epochs, but validation loss bottoms near epoch 8 and rises afterward even as training loss keeps falling, a mild overfitting signature. Validation mean IoU nonetheless keeps improving through epoch 14, which is why the best checkpoint is selected by that metric rather than final epoch or validation loss.

Grouping classes by ground truth support shows the rare class weakness directly. The three most frequent classes, background, agriculture and water, average 0.5822 mean IoU and 0.7449 mean recall. The two rarest, road and barren, average only 0.3946 and 0.4482. Figure 3 shows this alongside the confidence comparison. Boundary pixels, within three pixels of a class edge, reach only 0.3076 mean IoU against 0.5803 on interior pixels. Mean confidence is 0.8490 on correct pixels but still 0.7215 on incorrect ones, so the model stays fairly sure of itself even when wrong, exactly the situation a label free failure signal needs to help with.

B. Label free signals hold up across domains

Table II reports pixel level ranking quality for entropy and the blended failure score on all four splits, Figure 4 plots the same numbers alongside risk coverage. Entropy AUROC is 0.7094 in domain on LoveDA, rises to 0.7215 on Potsdam and 0.7371 on Vaihingen, then drops to 0.6091 on OpenEarthMap despite its base error rate sitting between the other two. The blended failure score follows the same shape, 0.6998, 0.7123, 0.7141 and 0.6037. Confidence shortfall and margin risk, not plotted, follow the identical trend within a few points of entropy. Every domain stays clearly above the 0.5 chance level, but the two single city European domains rank noticeably better than the worldwide one. On Potsdam, keeping only the fifth of pixels the failure score trusts most cuts the error rate from about 51 percent overall to under 10 percent among pixels kept.

Image level ranking is noisier, as expected with far fewer images per split than pixels. The Spearman correlation between an image's mean entropy and true error rate is 0.4175 on LoveDA, drops to 0.2280 on the 32 Potsdam

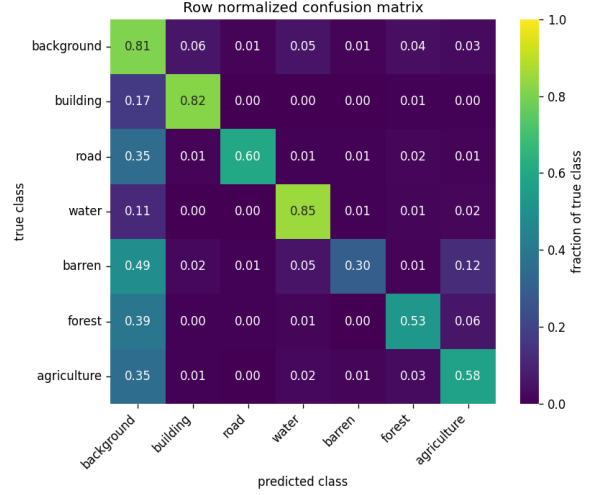
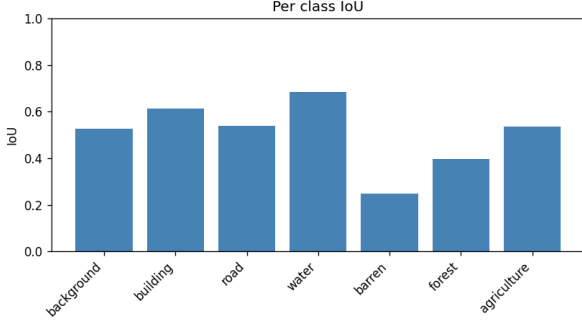


Fig. 1: Per class IoU (left) and confusion matrix (right) on LoveDA validation. Barren and forest are weakest, confused with agriculture and background most.

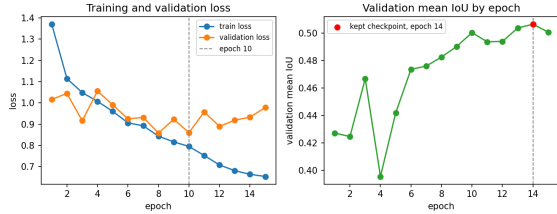


Fig. 2: Training and validation loss (left) and validation mean IoU (right) across 15 epochs. Validation loss bottoms near epoch 8 and rises while training loss keeps falling, yet mean IoU keeps improving through the kept checkpoint at epoch 14.

TABLE II: Pixel level ranking quality across four domains. Chance level for AUROC is 0.5. LoveDA is in domain, the other three are cross domain.

Domain	Base error rate	Entropy AUROC	Failure score AUROC
LoveDA	0.3118	0.7094	0.6998
Potsdam	0.5071	0.7215	0.7123
Vaihingen	0.5962	0.7371	0.7141
OpenEarthMap	0.5529	0.6091	0.6037

patches, rises to 0.3444 on the 49 Vaihingen patches and is weakest at 0.1987 on the 384 OpenEarthMap tiles. All four stay clearly positive.

C. Whole image novelty is not a reliable domain alarm

Table III reports the domain AUROC, whether the novelty score alone separates a foreign domain’s images from the training adjacent baseline distances. Vaihingen reaches 0.8172, a strong separation, OpenEarthMap follows at 0.7132, also clear despite being RGB imagery like Potsdam. Potsdam sits at 0.4018, below chance. Figure 5 shows all three histograms. The gap between Potsdam and OpenEarthMap rules out a sensor band change as the sole explanation, since neither differs from LoveDA in sensor type yet they land at opposite ends of the range. This inconsistency is the honest finding, a whole image novelty score separates some foreign domains well and others not at all, with no simple rule predicting which. Per image novelty still correlates positively with true error rate on every domain, 0.3398 on Potsdam, 0.6460 on Vaihingen and 0.4287 on OpenEarthMap, so it is not useless, it simply cannot be trusted alone as a first pass alarm.

D. Qualitative failure cases

Figure 6 shows two of the lowest scoring LoveDA validation images by mean IoU, each as input, ground truth, prediction and error overlay. Both show large contiguous misclassified regions rather than scattered boundary noise, consistent with the findings above, entire rare class regions mislabeled as a common neighbor class with high confidence.

E. Limitations and future work

The cross domain class mappings are coarse by construction, low vegetation maps to background rather than agriculture, car and clutter are excluded,

TABLE III: Whole image novelty as a domain alarm across three foreign domains.

Domain	Novelty domain AUROC	Novelty image Spearman
Potsdam	0.4018	0.3398
Vaihingen	0.8172	0.6460
OpenEarthMap	0.7132	0.4287

and OpenEarthMap’s rangeland and developed space both collapse onto background. Absolute error rates on the foreign domains should be read with this caveat, though the ranking metrics carrying the main claim only need the error mask roughly right. Potsdam and Vaihingen are small, 32 and 49 patches, so their image level Spearman correlations carry wide uncertainty. OpenEarthMap’s 384 tiles are a real improvement yet its Spearman is still the weakest, plausibly because scene diversity makes ranking harder. The backbone starts from ImageNet weights rather than remote sensing imagery, and training was capped at 384 pixels rather than LoveDA’s native 512 due to free tier cloud GPU limits, both likely capping the accuracy ceiling.

Next steps follow directly, training at native 512 resolution with a larger backbone such as ResNet101, pretraining on remote sensing imagery to close the domain gap, additional labeled benchmarks that vary ground sample distance and scene diversity independently, a better calibrated novelty detector, and interpreting failures with geographic information science expertise.

VII. CONCLUSION

A standard DeepLabV3 ResNet50 baseline on LoveDA reproduces published performance across two independently trained runs while carrying clear boundary, rare class and overconfidence weaknesses. Four label free pixel level uncertainty signals rank wrong pixels above correct ones with an AUROC that stays clearly above chance across the training domain and three foreign domains, extending a result previously shown in medical and driving imagery into remote sensing. Ranking quality is highest on the two single city domains and weaker on the worldwide one. A whole image Mahalanobis novelty score does not show the same consistency, swinging from below chance on Potsdam to above chance on Vaihingen and OpenEarthMap with no simple property predicting which.

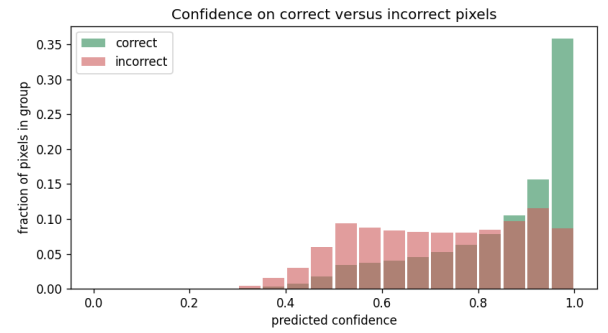
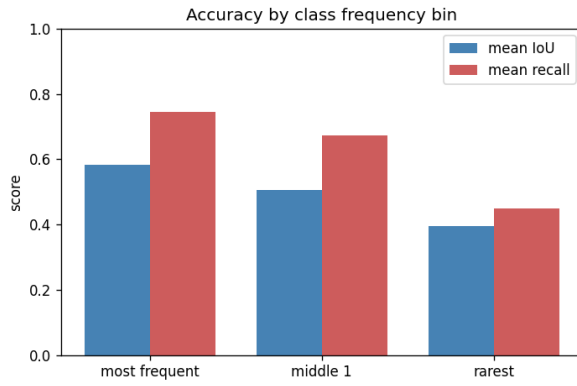


Fig. 3: Mean IoU by class frequency bin (left) and confidence on correct against incorrect pixels (right). Rare classes lag and the model stays confident on many of its errors.

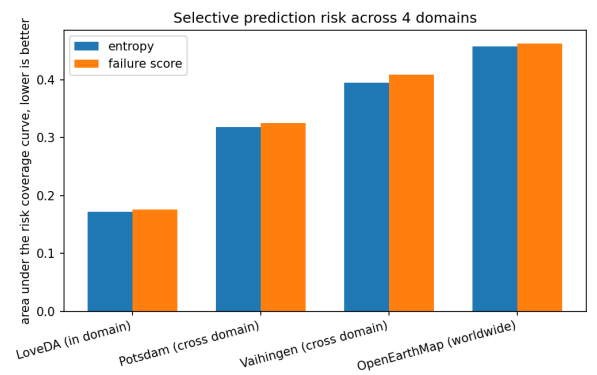
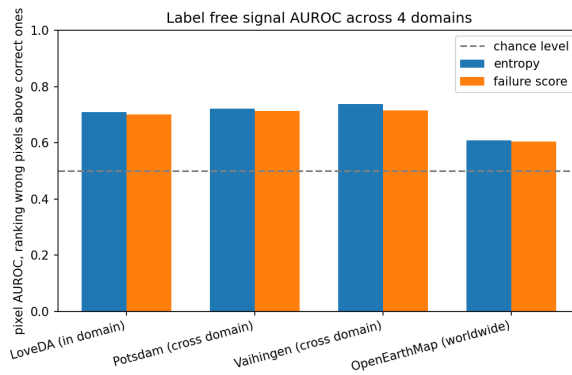


Fig. 4: Pixel level signal AUROC across LoveDA and the three foreign domains, dashed line marks chance (left), and area under the risk coverage curve, lower is better (right).

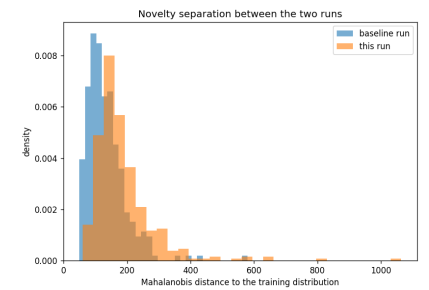
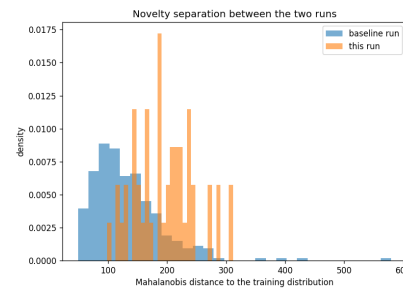
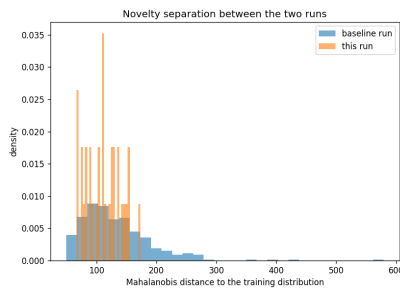


Fig. 5: Mahalanobis distance to the LoveDA training distribution, baseline against Potsdam (left), Vaihingen (middle) and OpenEarthMap (right). The three domains separate very differently despite two sharing the RGB sensor type.

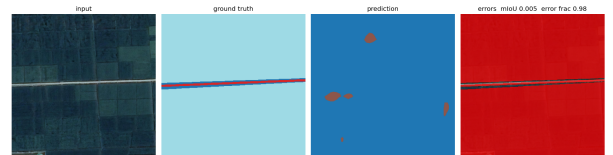
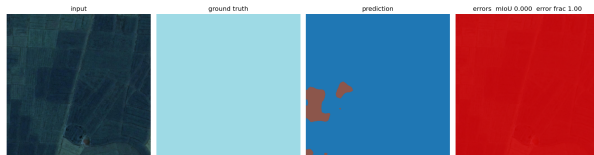


Fig. 6: Two of the lowest scoring LoveDA validation images, input, ground truth, prediction and error overlay from left to right within each panel.

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